# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – CONCEPT MAPPING

| | | | |
|---|---|---|---|
| **Course Code:** | CSA16 | **Course Title:** | Data Warehousing and Data Mining |
| **CO Assessed:** | CO1 – Precision (BL2, BL3, BL4) | **Assessment:** | Assessment Tool 3 – Concept Mapping |
| **Weightage:** | 20% of CIA | **Total Marks:** | 100 (5 Questions × 100 Marks) |
| **CO1 Statement:** | *Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2* | | |
| **Student Name:** | Hemalakshmi H | **Reg. No.:** | 192571049 |
| **Section / Batch:** | | **Date:** | 6/10/2026 |

**Instructions to Students**

- Answer the given question through a technical presentation. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled..

| Question Type | Focus Area | Questions | Marks |
|---|---|---|---|
| **Concept Mapping** | Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting. | Q1, Q2,Q3,Q4,Q5 | 100 |
| | **GRAND TOTAL:** | **100 Marks** | |

## SECTION A – Technical Presentation

### Concept Map 1: Decision Support Systems (DSS)

### Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.
Core components: Data, Model, User Interface

1. Types of DSS
2. Flow: Data → Analysis → Decision
3. Role in business decision-making
4. Real-world example linkage

## Concept Map 2: Operational Systems vs DSS

### Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

## Concept Map 3: Architecture of Data Warehouse

### Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

## Concept Map 4: OLAP, Data Cubes & Dimensional Modelling

### Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modelling techniques.

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

## Concept Map 5: Query Processing & BI Tools (KNIME)

### Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integration

**Code of Conduct**

*I Hemalakshmi H (192571049) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

## RUBRICS FOR CALCULATION

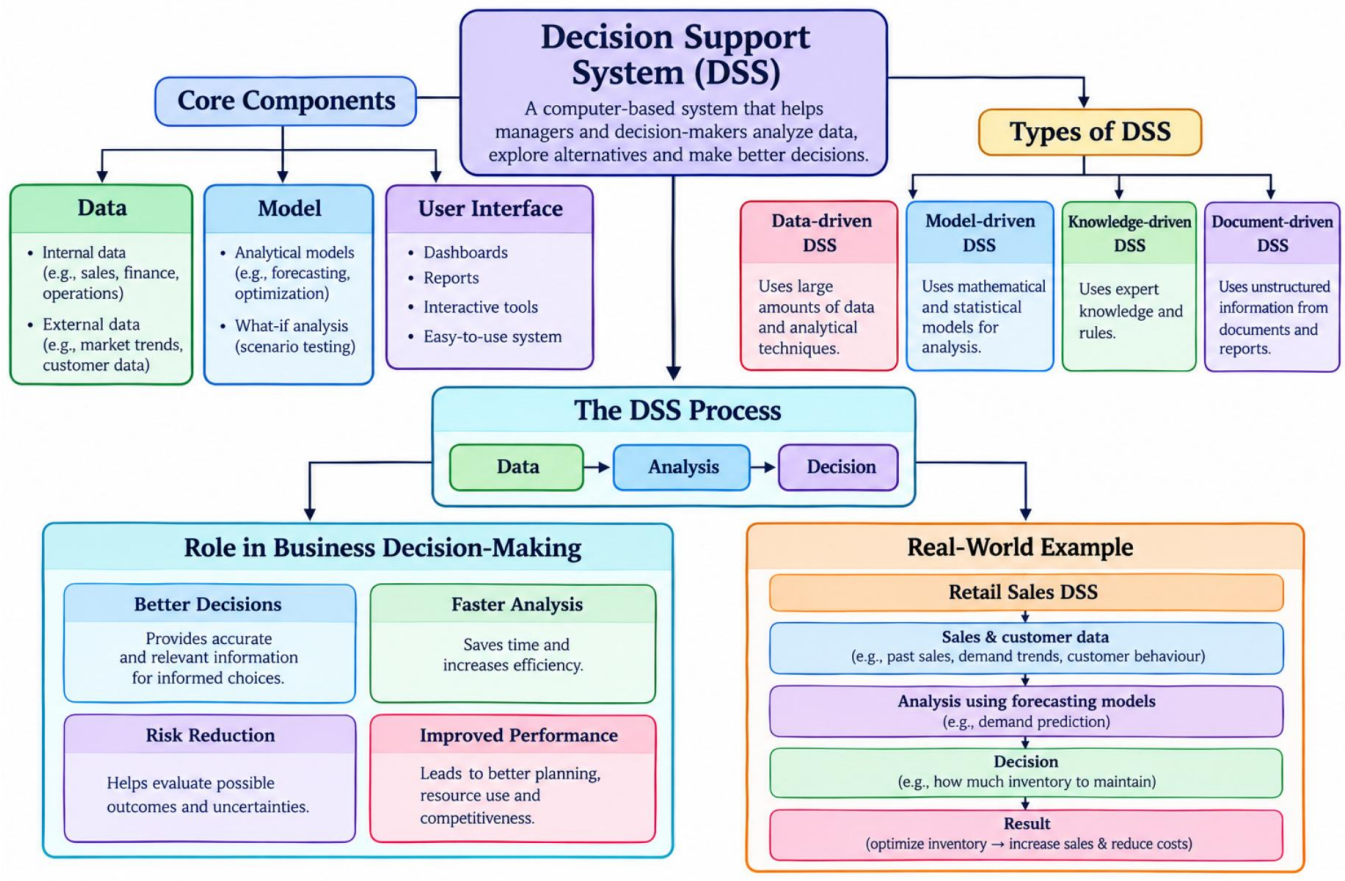
| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Concept Identification | 25% | Identifies all key concepts from literature accurately and comprehensively | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts |
| Linkages | 25% | Establishes clear, meaningful, and accurate relationships between concepts | Shows correct relationships with minor gaps | Relationships are weak, unclear, or partially incorrect | No meaningful relationships established |
| Organization | 20% | Concepts are well-structured hierarchically with logical flow and grouping | Mostly organized with minor structural issues | Some organization but lacks clear hierarchy | No logical organization or structure |
| Integration of Literature | 20% | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature | Limited integration; mostly isolated concepts | No integration of literature evidence |
| Clarity | 10% | Concept map is clear, neat, visually structured, and easy to interpret | Generally clear with minor readability issues | Clarity is reduced; difficult to interpret in parts | Poorly presented and hard to understand |

## Marks Evaluation:

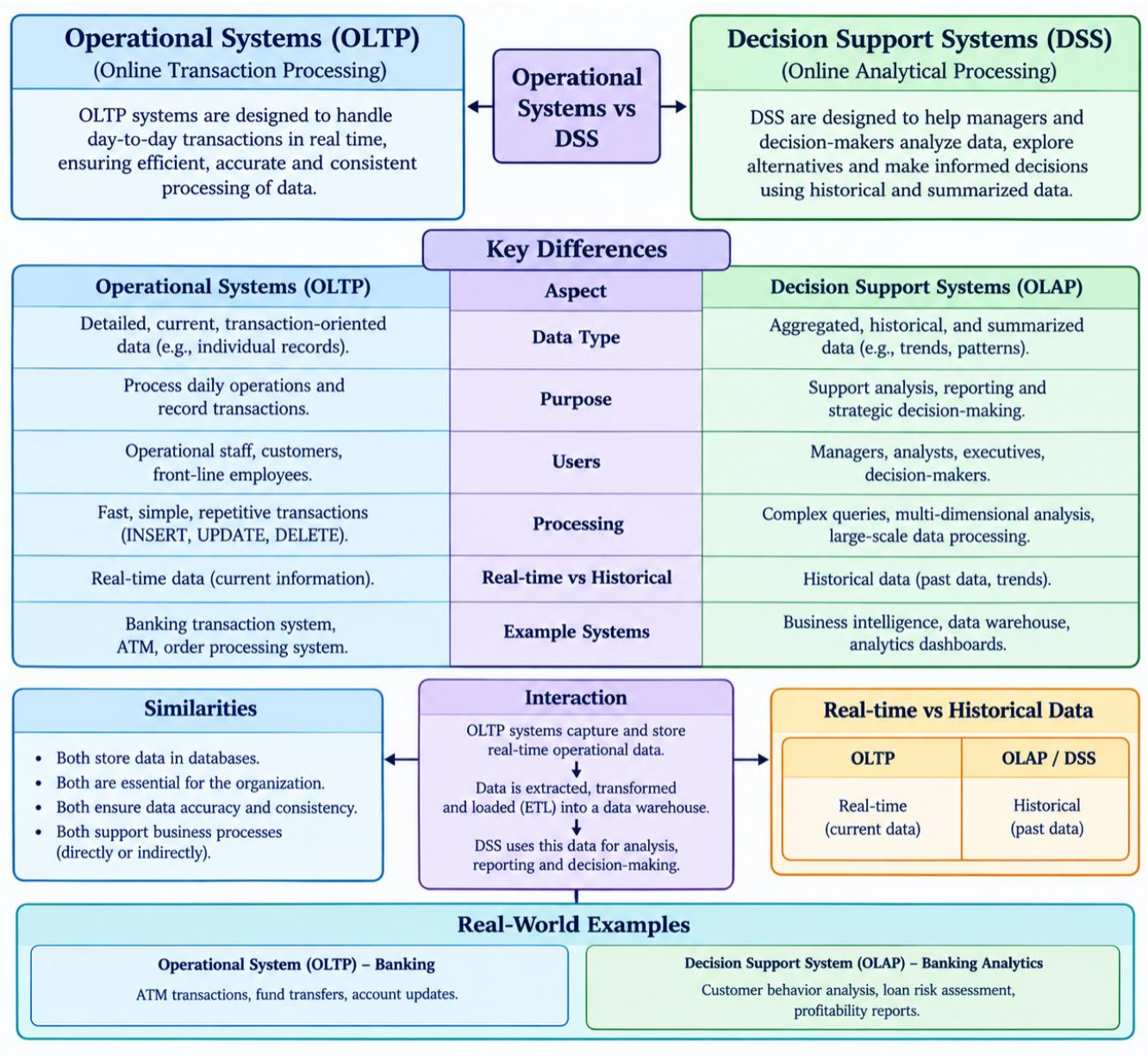
| Question No. | Concept Identification (25%) | Linkages (25%) | Organization (20%) | Integration of Literature (20%) | Clarity (10%) | Total Marks (Each – 50) |
|---|---|---|---|---|---|---|
| Q1 | | | | | | |
| Q2 | | | | | | |
| Q3 | | | | | | |
| Q4 | | | | | | |
| Q5 | | | | | | |
| | | | | | Overall Total (100) | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| | | |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |

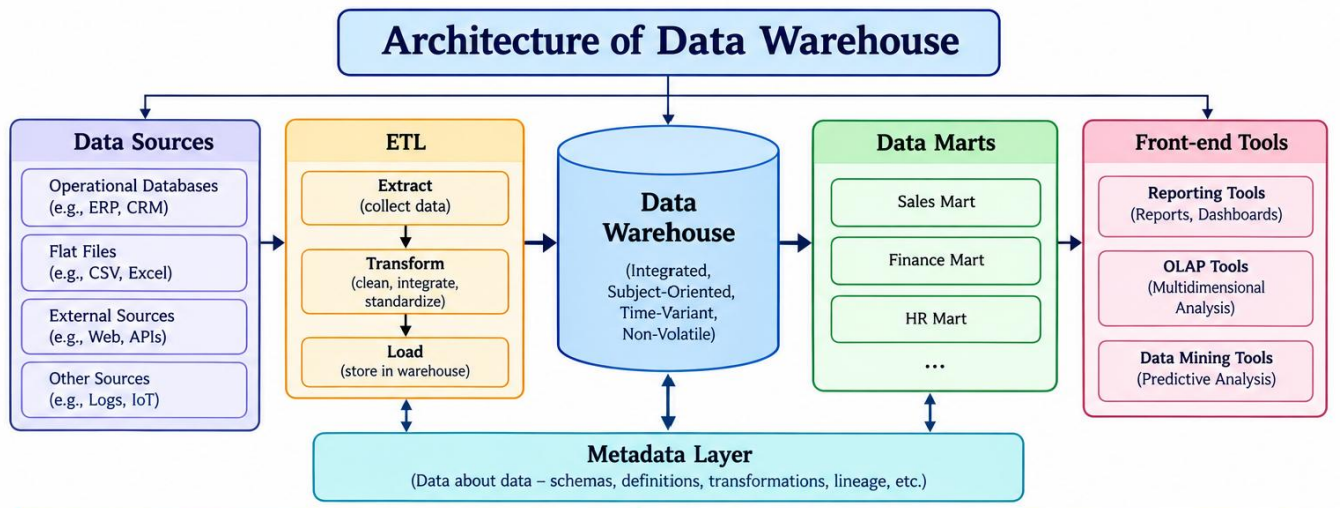
## Concept Map 1: Decision Support Systems (DSS)



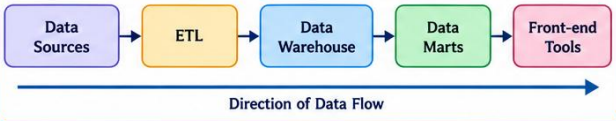
## Concept Map 2: Operational Systems vs DSS



## Concept Map 3: Architecture of Data Warehouse




### 1. Data Flow




### 2. Role of Each Layer

| Layer | Role |
|---|---|
| Data Sources | Provide raw data from internal and external sources. |
| ETL | Extracts, transforms and loads data into the warehouse. |
| Data Warehouse | Stores integrated, subject-oriented, historical data. |
| Data Marts | Provides department-specific data for faster access and analysis. |
| Metadata Layer | Manages data definitions, rules, relationships and lineage. |
| Front-end Tools | Enable reporting, visualization and decision-making. |

### 3. Key Similarities

- Both store data for analytical and business use.
- Both support decision-making.
- Both handle large volumes of data.
- Both work together (OLTP provides data, DSS analyzes it).

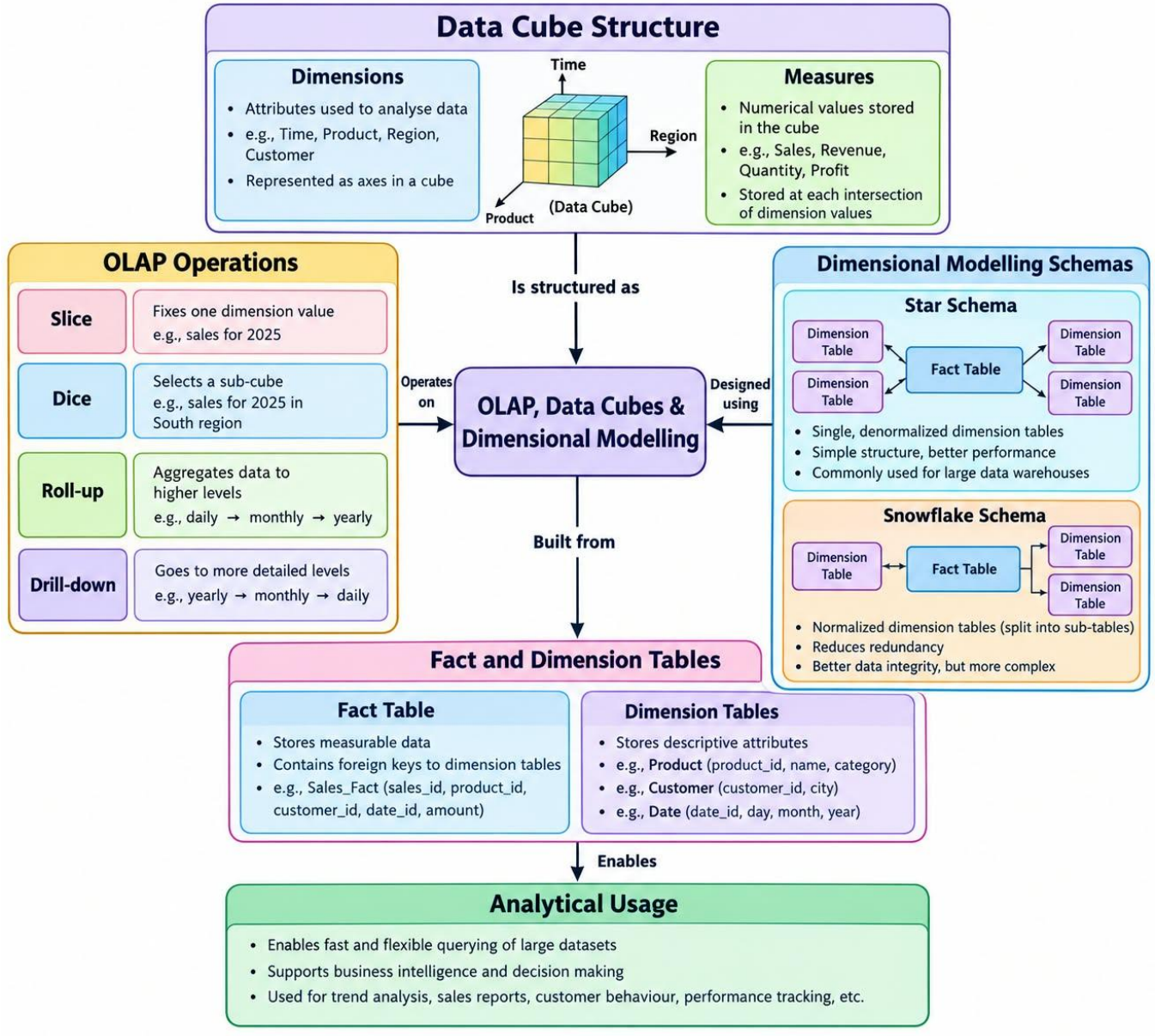
### 4. Real-time vs Historical Data

- **OLTP:** Real-time / current data (e.g., latest transactions).
- **DSS (OLAP):** Historical data (e.g., trends over months/years).

### 5. Example Systems

- **OLTP (Banking):** Core banking system, ATM transactions, account updates.
- **DSS (Analytics):** Business intelligence tools (e.g., Tableau, Power BI), data warehouse reporting.

## Concept Map 4: OLAP, Data Cubes & Dimensional Modelling



## Concept Map 5: Query Processing & BI Tools (KNIME)

